

GRINNELL, IA, USA

WMO: 725292

Lat: 41.717N

Lon: 92.700W

Elev: 1008

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 14976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-0.7	-12.9	2.8	-4.8	-8.3	3.7	0.6	31.9	20.2	28.5	19.9	9.3	310	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	90.8	74.8	88.2	73.6	85.9	72.4	78.1	86.4	76.3	84.6	74.6	82.9	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	138.5	84.2	73.3	128.5	81.4	72.1	123.3	80.2	42.4	85.8	40.3	84.7	38.8	83.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	23.5	20.1	-14.2	94.5	6.1	3.4	-18.6	96.9	-22.2	98.9	-25.6	100.8	-30.1	103.3
			-14.7	81.2	5.9	1.5	-19.0	82.2	-22.4	83.1	-25.7	83.9	-30.0	85.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	20.8	24.2	37.5	49.9	62.0	71.4	74.2	71.6	65.7	52.5	38.9	27.0
	DBStd	21.03	12.70	12.31	11.97	9.70	8.72	5.51	5.52	5.31	7.80	9.11	10.45	10.85
	HDD50	3320	905	725	415	118	11	0	0	0	2	77	354	713
	HDD65	6452	1370	1143	853	458	162	10	4	7	85	399	784	1177
	CDD50	3244	0	2	29	116	383	642	751	671	473	156	20	1
	CDD65	902	0	0	2	6	69	202	290	213	107	13	0	0
	CDH74	7831	0	0	18	95	628	1742	2690	1673	867	117	1	0
	CDH80	2477	0	0	1	16	179	571	948	481	259	22	0	0
Wind	WSAvg	9.3	10.9	10.7	10.9	11.9	10.4	8.4	6.5	5.9	7.1	8.9	10.6	10.1
Precipitation	PrecAvg	37.2	1.1	1.3	2.3	3.8	4.8	5.5	4.4	4.8	3.2	2.7	2.0	1.4
	PrecMax	53.6	2.2	3.0	5.1	7.7	10.3	12.2	11.6	13.1	7.1	6.3	5.4	4.4
	PrecMin	26.6	0.5	0.2	0.2	1.4	0.9	0.9	0.5	0.8	0.9	0.5	0.1	0.0
	PrecStd	8.1	0.4	0.8	1.5	1.8	1.9	2.6	2.7	3.1	1.6	1.9	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	62.8	76.8	82.2	90.6	91.6	95.7	91.8	90.9	82.5	71.5	57.5
		MCWB	45.1	52.5	60.6	62.9	70.7	74.6	77.2	73.9	73.3	66.0	57.1	53.4
	2%	DB	44.9	52.2	68.0	76.6	84.7	89.6	91.0	88.3	86.8	77.1	64.1	50.4
		MCWB	40.1	45.5	57.0	60.1	66.8	74.1	76.6	74.8	72.2	63.0	54.7	46.0
	5%	DB	40.7	45.2	61.3	71.6	80.9	86.5	88.1	85.5	82.5	72.4	58.9	45.8
		MCWB	37.4	40.7	52.9	57.3	65.1	71.0	75.4	73.6	69.8	61.6	50.2	42.2
	10%	DB	36.9	39.9	54.8	66.5	77.0	83.2	85.6	82.0	79.3	68.1	54.2	40.6
		MCWB	34.8	36.7	47.6	54.7	63.6	69.5	73.7	71.6	67.2	58.7	47.5	37.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	53.8	63.4	66.5	73.3	79.4	81.2	79.3	76.0	70.9	59.9	53.5
		MCDB	52.3	60.9	72.7	77.9	84.1	88.9	89.4	88.2	85.8	79.1	66.5	55.3
	2%	WB	40.6	46.2	59.3	62.8	70.4	76.4	79.0	76.9	74.2	67.4	55.9	47.5
		MCDB	44.3	51.5	64.4	72.1	80.7	85.6	86.8	84.8	83.4	72.2	63.2	49.0
	5%	WB	37.5	41.2	53.4	59.8	68.4	74.1	77.0	75.0	72.2	63.7	51.7	41.7
		MCDB	40.2	45.3	59.9	69.5	77.8	82.9	84.9	82.7	80.1	69.4	57.6	45.5
	10%	WB	34.9	36.8	48.2	56.6	66.2	72.2	75.1	73.4	70.1	59.9	48.0	37.4
		MCDB	36.9	39.9	54.4	64.8	73.8	80.4	83.1	80.6	76.3	67.0	53.3	39.8
Mean Daily Temperature Range	5% DB	MDBR	15.3	15.8	16.9	20.4	19.7	18.7	18.5	18.4	20.6	19.6	17.4	14.2
		MCDBR	18.6	21.9	26.2	27.9	23.7	21.9	20.6	20.9	22.6	24.8	24.2	19.5
		MCWBR	14.7	15.9	15.5	15.0	11.1	10.2	9.3	10.2	10.2	12.8	15.4	14.9
	5% WB	MCDBR	17.8	22.0	23.0	24.2	20.4	19.3	18.1	17.8	19.2	19.8	22.2	19.0
		MCWBR	14.4	16.4	14.8	15.2	10.9	11.2	9.8	9.8	9.8	13.6	16.1	15.3
Clear-Sky Solar Irradiance	taub	0.286	0.305	0.330	0.378	0.408	0.419	0.435	0.414	0.383	0.339	0.299	0.285	
	taud	2.408	2.360	2.388	2.275	2.262	2.280	2.217	2.293	2.332	2.452	2.492	2.430	
	Ebn at Noon	275	286	291	283	276	272	266	268	269	271	269	265	
	Edh at Noon	25	31	34	41	43	42	44	40	36	28	23	23	
All-Sky Solar Radiation	RadAvg	603	927	1209	1499	1765	1953	1977	1719	1451	963	652	495	
	RadStd	58	67	117	115	128	192	137	105	107	113	56	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page